MIL-E-1/835B
5 February 1969
SUPERSEDING
MIL-E-1/835A
29 February 1960

MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, MAGNETRON

## TYPE 6510

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Fixed frequency, 9,375 ±30 MHz, pulse type, integral magnet, rated peak power output 80 kw.

ABSOLUTE RATINGS:

| Parameter: | Ef | If(surge) | tk | tpc | ib | Pi | T(anode) | VSWR | T(cathode bushing) | Alt |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | a | sec | μs | a | W | °C | --- | °C | ft |
| Maximum: | 14 | 10 | --- | 5.0 | 16.5 | 240 | 150 | 1.5:1 | 175 | 12,000 |
| Minimum: | --- | --- | 120 | 0.4 | --- | --- | -55 | --- | --- | --- |

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Magnet isolation: 2 inches
Coupling: Mates with UG-51/U-RG-51/U or UG-138/U-RG-68/U

TEST CONDITIONS:

| Parameter: | Ef | tpc | Du | rrv | Ib | VSWR |
|---|---|---|---|---|---|---|
| Unit: | V | μs | --- | kv/μs | mAdc | --- |
| Tolerance: | --- | ±10% | --- | min | --- | max |
| Test 1: | 9.1 | 0.4 | 0.00065 | 160 | 9.8 | 1.05 |
| Test 2: | 7.8 (see notes 3 and 4) | 5.0 | 0.001 | 100 | 15 | 1.05 |

GENERAL:

Qualification - Required



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| METHOD | REQUIREMENT OR TEST | NOTES | TEST | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MINIMUM | MAXIMUM | |
| | Quality conformance inspection, part 1 | 1, 16 | --- | | | | | |
| 1301 | Heater current | --- | --- | Ef = 12.6 V; tk = 120 seconds (min) | If | 2.0 | 2.4 | A |
| --- | Fixed frequency | --- | 2 | T (anode) = 100° ±10° C | F | 9,345 | 9,405 | MHz |
| 4003 | Pressurization | --- | --- | 40 to 45 psia input and output assemblies | --- | --- | --- | --- |
| 4306 | Pulse voltage | --- | 2 | | epy | 14 | 16 | kv |
| 4250 | Power output | --- | 2 | | Po | 70 | --- | W |
| 4308 | RF bandwidth | 15, 17 | 1 | Ib = 8.8 and 10.8 mAdc | BW | --- | $\frac{2.0}{tpc}$ | MHz |
| | | 15, 17 | 2 | Ib = 13.5 and 16.5 mAdc | BW | --- | $\frac{2.5}{tpc}$ | MHz |
| 4308 | Minor lobes | 15, 17 | 1 | Ib = 8.8 and 10.8 mAdc | Ratio | 8 | --- | db |
| | | 15, 17 | 2 | Ib = 13.5 and 16.5 mAdc | Ratio | 6 | --- | db |
| 4310 | Frequency pulling factor | --- | 1 | | ΔF | --- | 13 | MHz |
| 4311 | Frequency pushing factor | --- | 2 | Ib = 13.5 to 16.5 mAdc | $\frac{\Delta F}{mA}$ | --- | 1 | MHz/mA |
| 4315 | Stability | 10, 11, 13 | 1 | | MP | --- | 0.25 | % |
| | | 10, 11, 13, 21 | 2 | | MP | --- | 0.25 | % |
| 4318 | Jitter measurements | 18 | --- | | | | | |
| | Time jitter | 19 | 1 | | Δtj | --- | 0.0015 | μs |
| | Frequency jitter | 20 | 1 | | ΔFj | --- | 0.055 | MHz |
| | Amplitude jitter | --- | 1 | | ΔAj | --- | 0.05 | db |
| | Quality conformance inspection, part 2 | 2, 16 | --- | | | | | |
| --- | Vibration | 6 | --- | Heater voltage only; Ef = 12.6 V; 15G; F = 60 Hz; duration = 15 minutes | --- | --- | --- | --- |
| --- | Shelf life | 22 | --- | t = 90 days | --- | --- | --- | --- |
| --- | Shelf-life end points | | | | | | | |
| 4315 | Stability | 10, 12, 13 | 2 | | MP | --- | 0.5 | % |
| 4315 | Stability | 18 | 2 | | MP | --- | 10.0 | % |
| 4315 | Stability | 10, 11, 13 | 2 | rrv = 60 kv/μs (max) | MP | --- | 0.25 | % |



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| METHOD | REQUIREMENT OR TEST | NOTES | TEST | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MINIMUM | MAXIMUM | |
| | Quality conformance inspection, part 3 | 16 | --- | | | | | |
| --- | Life test provisions | 7, 23, 24, 28 | --- | Group D | --- | 1,666 | --- | Cycles |
| --- | Life test end points | | | | | | | |
| 4250 | Power output | 25 | 2 | | Po | 56 | --- | W |
| --- | Fixed frequency | --- | 2 | | F | 9,345 | 9,405 | MHz |
| 4308 | RF bandwidth | 15, 17, 26 | 2 | | BW | --- | $\frac{2.5}{tpc}$ | MHz |
| 4308 | Minor lobes | 15, 17, 26 | 2 | | Ratio | 6 | --- | db |
| 4315 | Stability | 10, 11, 13, 26 | 1 | | MP | --- | 2 | % |
| | | 10, 11, 13, 26 | 2 | | MP | --- | 2 | % |
| 4318 | Jitter measurements | 18 | --- | | | | | |
| | Time jitter | 19 | 1 | | $\Delta tj$ | --- | 0.0017 | $\mu s$ |
| | Frequency jitter | 20 | 1 | | $\Delta Fj$ | --- | 0.060 | MHz |
| | Amplitude jitter | | 1 | | $\Delta Aj$ | --- | 0.055 | db |
| | Periodic-check tests | 16, 27 | --- | | | | | |
| --- | Vibration | 5, 6, c | 1 | 2G | $\Delta F$ | --- | 5.0 | MHz |
| 1041 | Shock | --- | --- | No voltages; 15 G for 11 ms duration | --- | --- | --- | --- |
| --- | Input capacitance | --- | --- | Measured between cathode terminal and mounting plate | Cin | 10.0 | 14.0 | pF |
| 4309 | Magnetron nonoscillating impedance (phase of sink) | 8 | 1 | | Dist | 0.26 | 0.40 | $\lambda g$ |
| 1002 | Barometric pressure, reduced | 9 | 2 | Pressure = 500 mm Hg absolute | --- | --- | --- | --- |
| 4027 | Temperature coefficient | v | 1 | T(anode) = 70° to 100° C | $\frac{\Delta F}{\Delta °C}$ | --- | 0.25 | MHz/°C |
| 1047 | Low-temperature operation | --- | --- | tk = 120 sec (max) | --- | --- | --- | --- |
| 4315 | Stability | 10, 11, 13 | 1 | | MP | --- | 1 | % |
| 4315 | Stability | 10, 12, 13 | 1 | | MP | --- | 5 | % |
| 1143 | Forced convection | 14, v | 1 | TA = 25° C (approximate) | $\Delta T$ | --- | 70 | °C |



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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1 percent, inspection level II.

2. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 2, shall be 6.5 percent, inspection level S3.

3. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4, 000 pF across the heater directly at the input terminals.

4. A tk of 120 seconds (maximum) with an Ef of 12.6 volts is required before test.

5. F = 50 to 500 to 50 Hz in not less than 5 minutes at logarithmic rate.

6. Direction of vibration shall be in a plane perpendicular to the axis of the cathode. There shall be no heater-cathode or cathode-anode shorts during test.

7. The tube shall be cycled in accordance with the following schedule:

| Condition | | Duration |
|---|---|---|
| Standby | Ef = 12.6 V | 2 minutes |
| Test 1 | | 3 minutes |
| Test 2 | | 15 minutes |
| Off | | 10 minutes |

8. Using a standard cold-test technique, the phase of sink as measured from the output flange to the first minimum, toward the load, shall be within the limits specified.

9. The tube shall be operated in a transmission line with a load VSWR of 1.5:1 varied through all phases. At specified pressure, there shall be no evidence of breakdown at either the input or output assemblies.

10. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percent of the number of input pulses applied during the period of observation. The missing pulses (MP), due to any causes, are considered to be "missing" if the rf energy is less than 70 percent of the normal energy level in the frequency range of 9, 330 to 9, 425 MHz.

11. The missing pulses shall be counted during any consecutive 5 minute interval of a 10-minute test period.

12. The missing pulses shall be counted during the first minute of operation.

13. The tube shall be operated into a transmission line with a VSWR of 1.5:1 adjusted in phase to produce maximum instability.

14. An airflow of 15 cfm, at approximately 760 mm Hg and approximately 27° C shall be directed on the cooling fins from an orifice of 2-1/2 by 1-3/16 inches.

15. The tube shall be operated into a transmission line with a VSWR of 1.5:1 adjusted in phase to produce maximum spectrum degradation.

16. A modulator of the discharging network type shall be used for tests performed under test conditions 1 or 2.



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NOTES: -Continued

17. A suitable spectrum shall be considered one in which the major lobe has a shape such that its slope does not change sign more than once for power levels greater than the specified db below its peak.

18. The jitter stability limits are expressed as the root-mean-square (rms) value of the distribution of the parameter involved. A minimum of 10, 000 pulses shall be sampled to establish compliance with the specified limits.

19. Time jitter (tj) is defined as the jitter of the front edge of the rf pulse, at the 50-percent point, relative to the time of firing of the modulator.

20. Frequency jitter (Fj) shall be measured at the point of worst jitter within the rf pulse.

21. This test shall be the first test performed after the holding period.

22. This test shall be performed on four tubes per month when tube is in continuous production, but shipments of that month's production shall not be held pending completion of the test. So long as three of the four tubes for each of the first 3 months of a production run pass the test and 75 percent of the cumulative quantity of tubes tested pass the test, tubes shall be considered to conform to this specification. If either of the conditions are not met, shipments shall be halted until three or four tubes of current production conform to test.

23. Air cooling shall be adjusted so that the anode block runs at 150° C or at the maximum temperature it will reach in the absence of cooling, whichever is lower. This shall be adjusted during the test condition 2 portion of the cycle.

24. Prior to the performance of the life tests, the qualification, shock, and vibration tests shall be performed on the sample.

25. The cycling life-test power output for end of life shall be 70 percent of the start of life power or 50 watts whichever is the higher.

26. If during life test the tube does not meet the specified limits, it shall be recycled for an additional 5 cycles. At such time, the tests shall be repeated. Should the tube fail the second test, it shall be considered unsatisfactory.

27. Periodic-check tests are required if they have not been performed within the last 3 years, or there have been changes in design, material, or production methods.

28. VSWR = 1.5 minimum with phase varing through a minimum of $1/2\lambda$ approximately every 15 minutes.



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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2344)



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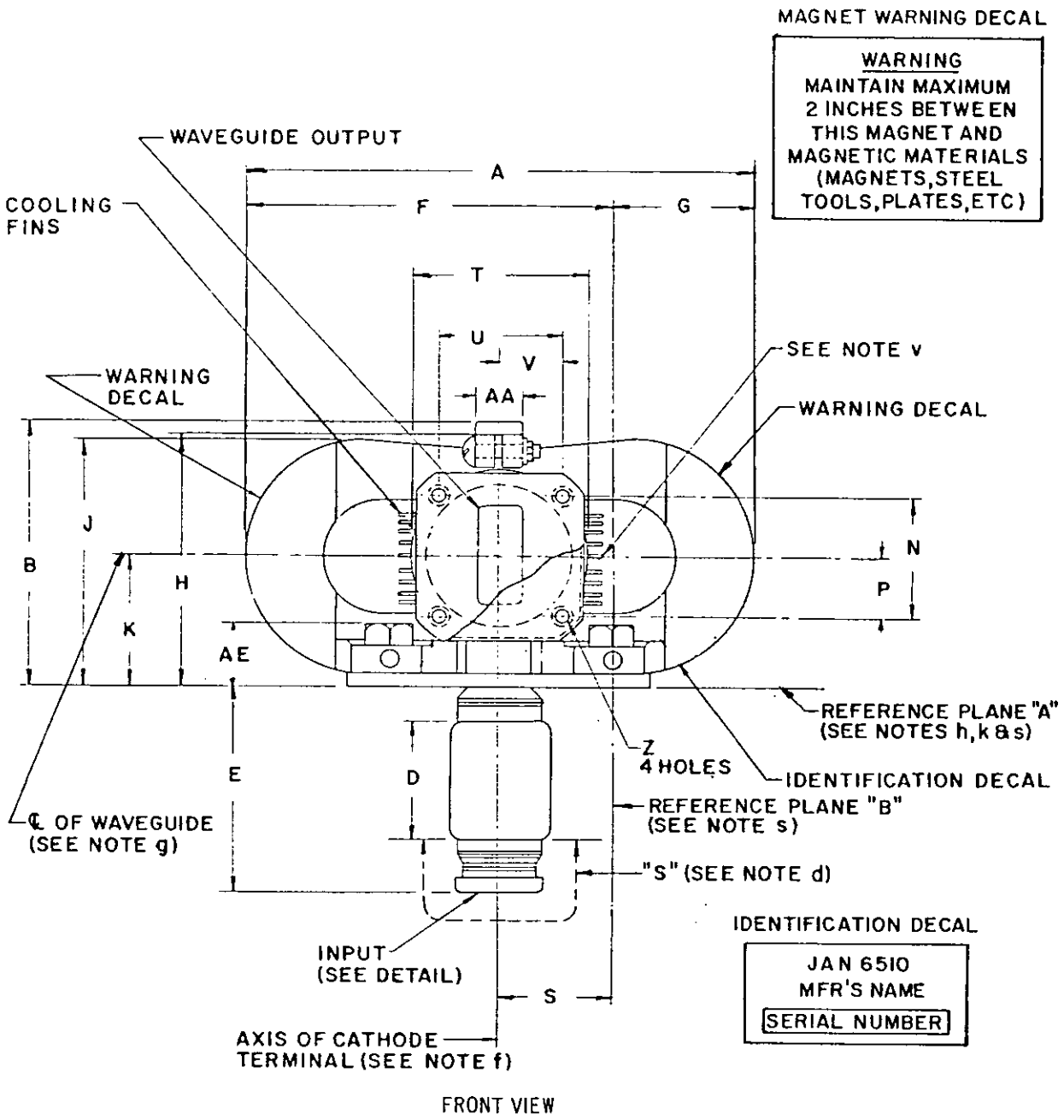
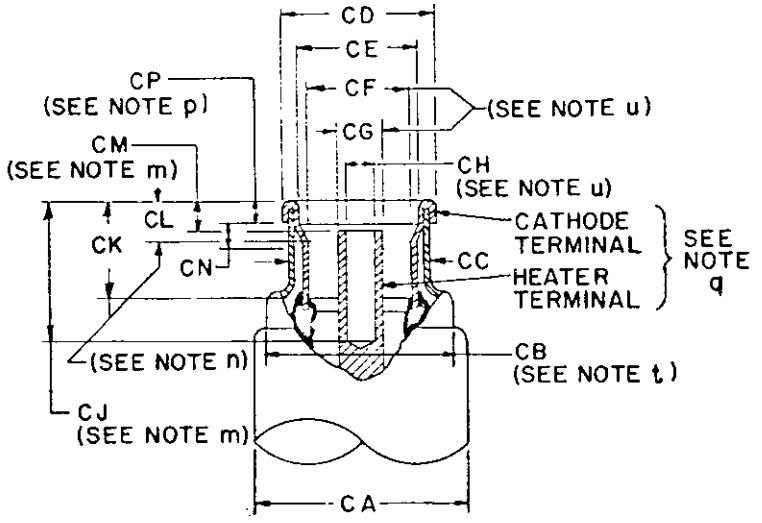



FIGURE 1. Outline drawing of electron tube type 6510.



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DETAIL OF INPUT CONNECTOR

FRONT VIEW & DETAIL DIM

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Notes |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 1 | | | |
| E | 2.094 (53.19) | 2.219 (56.36) | |
| K | 1.380 (35.05) | 1.426 (36.22) | |
| N | 1.348 (34.24) | 1.356 (34.44) | |
| P | .671 (17.04) | .681 (17.30) | |
| S | 1.250 (31.75) | 1.280 (32.51) | |
| U | 1.470 (37.34) | 1.478 (37.54) | |
| V | .732 (18.59) | .742 (18.85) | |
| CA | | 1.188 (30.18) | |
| CB | .995 (25.27) | 1.005 (25.53) | t |
| CC | .745 (18.92) | .755 (19.18) | |
| CD | .825 (20.96) | .830 (21.08) | |
| CE | .645 (16.38) | .655 (16.64) | |
| CF | .532 (13.51) | .545 (13.84) | |
| CG | .234 (5.94) | .266 (6.76) | |
| CH | .164 (4.17) | .174 (4.42) | |
| CJ | .750 (19.05) | | |
| CK | .516 (13.08) | | |
| CL | | .200 (5.08) | n |
| CM | .125 (3.18) | .187 (4.75) | m |
| CN | .109 (2.77) | .141 (3.58) | n |
| CP | .115 (2.92) | .135 (3.43) | p |

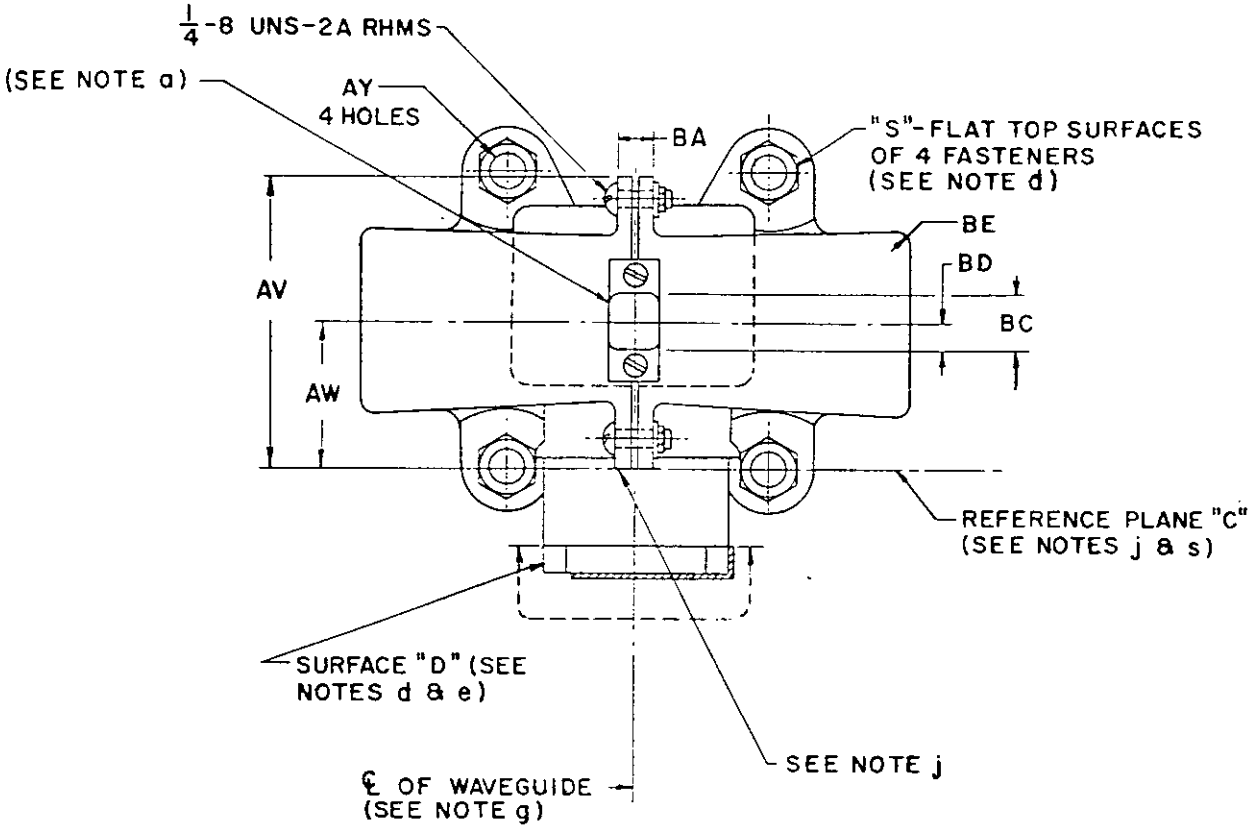
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | | 5.938 (150.83) |
| B | | 3.156 (80.16) |
| F | | 4.250 (107.95) |
| G | | 1.688 (42.88) |
| H | | 2.938 (74.63) |
| J | | 2.844 (72.24) |
| T | | 1.906 (48.41) |
| Z | No. 8-32 UNC-2B | |
| AA | | 1.438 (36.53) |
| AE | .594 (15.09) | .656 (16.66) |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Note |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 3 (periodic check) | | | w |
| D | 1.250 (31.75) | | |

FIGURE 1. Outline drawing of electron tube type 6510 - Continued.



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TOP VIEW

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 | | |
| AW | 1.344 (34.14) | 1.438 (36.53) |

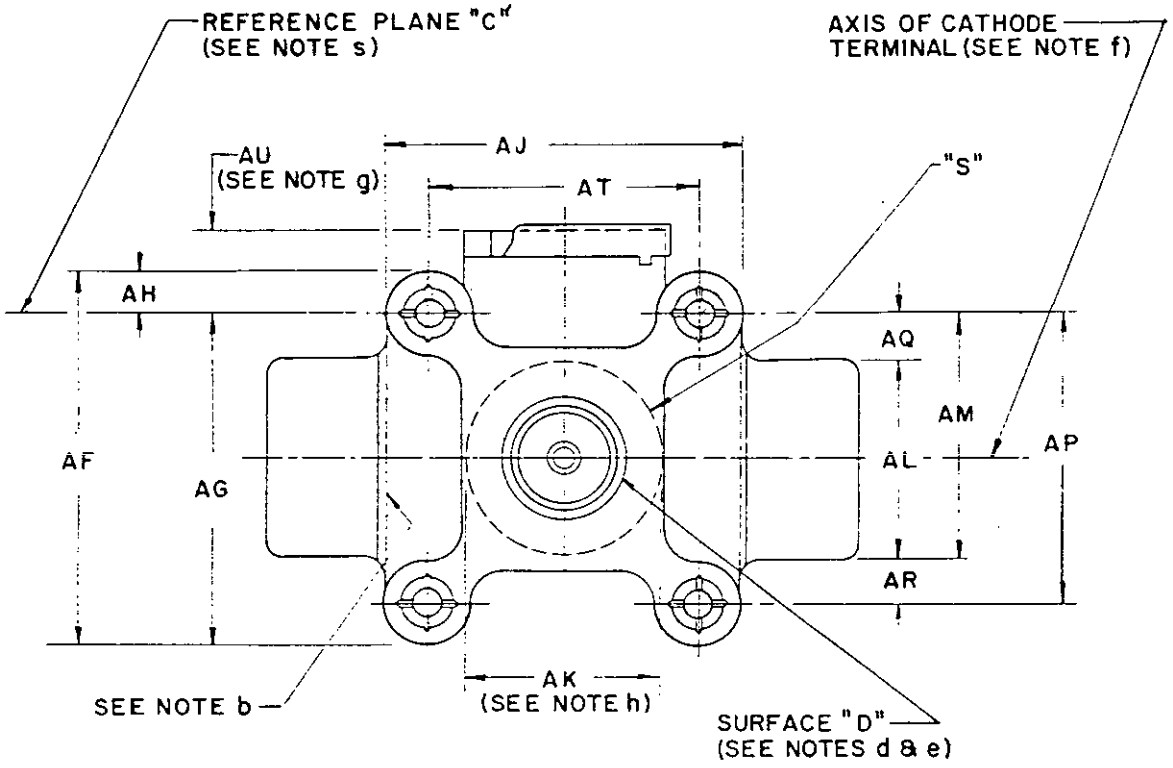
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| AV | | 2.875 (73.03) |
| AY | .276 (7.01) | .286 (7.26) |
| BA | | .625 (15.38) |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Note |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 3 (periodic check) | | | w |
| BC | | .813 (20.65) | |
| BD | | .406 (10.31) | |
| BE | | .031 R (.79) | |

FIGURE 1. Outline drawing of electron tube type 6510 - Continued.



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BOTTOM VIEW

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Note |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 1 | | | |
| AP | 2.771 (70.38) | 2.791 (70.89) | |
| AU | .990 (25.15) | 1.040 (26.42) | g |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| AF | | 3.625 (92.08) |
| AG | | 3.203 (81.36) |
| AH | | .422 (10.72) |
| AJ | | 3.438 (87.33) |
| AK | 1.875 (47.63) | |
| AL | | 1.938 (49.23) |
| AM | | 2.375 (60.33) |
| AQ | | .594 (15.09) |
| AR | | .594 (15.09) |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Note |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 3 (periodic check) | | | w |
| AT | 2.521 (64.03) | 2.541 (64.54) | |

FIGURE 1. Outline drawing of electron tube type 6510 - Continued.



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NOTES:

a. Seal-off protector may be round in shape.
b. Alternate shape of mounting plate.
c. For vibration tests, planes of vibration shall be reference planes A, B and C.
d. All metal surfaces black-finished except those designated S and D. Surfaces designated S shall be gold, silver, nickel, or bright alloy plated, or brass surfaces except as specified.
e. Hermetic connections may be made to surface D.
f. Axis of cathode terminal shall occupy specified locationwithin .046 (1.17 mm) R. Note g shall apply.
g. Limits include angular and lateral deviations.
h. With diameter AK resting on a plane surface coincident with reference plane A, a gage .010 (.25 mm) thick and .125 (3.18 mm) wide shall not enter, and areas of the base plate outside diameter AK shall be within .010 (.25 mm) of the plane surface.
j. Extremity of magnet lug shall coincide with reference plane C within .094 (2.39 mm).
k. Any portion of the assembly which extends below reference plane A shall be within .625 (15.88 mm) R of specified axis
m. Defines extremity of cylindrical section designated by .169 (4.29 mm) diameter.
n. Defines extremity of cylindrical section designated by .540 (13.72 mm) diameter.
p. Clamping shall be confined to this area.
q. Heater terminal and cathode terminal shall be concentric within .010 (.25 mm)
r. Protective covers shall be provided with tube.
s. Reference planes A, B and C are mutually perpendicular.
t. Reference point for cathode temperature measurements located on diameter CB.
u. Gold, silver or nickel plate over full extent of maximum plug penetration.
v. Reference point for anode temperature measurements located where a central fin meets the anode block near the output section.
w. Periodic-check tests are required if they have not been performed within the last 3 years, or there have been changes in design, material or production methods.

FIGURE 1. Outline drawing of electron tube type 6510 - Continued.



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